1B7: Simulation of planets orbits

University Of Oslo
AST2000

Candidate : 15802

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Abstract

In this report, we try to simulate the orbits of planets in an 8-planet solar system. We will focus on the two-body problem, using both analytical and numerical methods to achieve this.

1 Introduction

This report's main goal is to understand how planets move around a central star and to solve the two-body problem to simulate these orbits. We will make some simplifications : for example, we'll ignore :the forces between planets because they're so far apart and the force a planet puts on the central star due to the vast difference in their masses. Our analytical method is based on basic physics principles, while our numerical method will use the Euler-Cromer method to solve Newton's second law as a second-order differential equation.

2 Method

The analytical solution :

We have found in the lectures note 1B7(1) ,the elliptical orbit can be expressed in terms of distance r as:

$$r = \frac{a(1-e^2)}{1+e\cos f} \tag{1}$$

Where :

- a is the semi major axis of the orbit
- e is the eccentricity which determines how the orbit will deviate from a prefect circle .It takes the values $0 < e < 1$ for an ellipse orbit.
- f is the true anomaly $f = \theta - \omega$ and is defined as the angle between the planet and the perihelion .
- θ the angle between the position vector and x-axis of a fixed coordinate system.
- ω is the aphelion angle

As shown in figure1

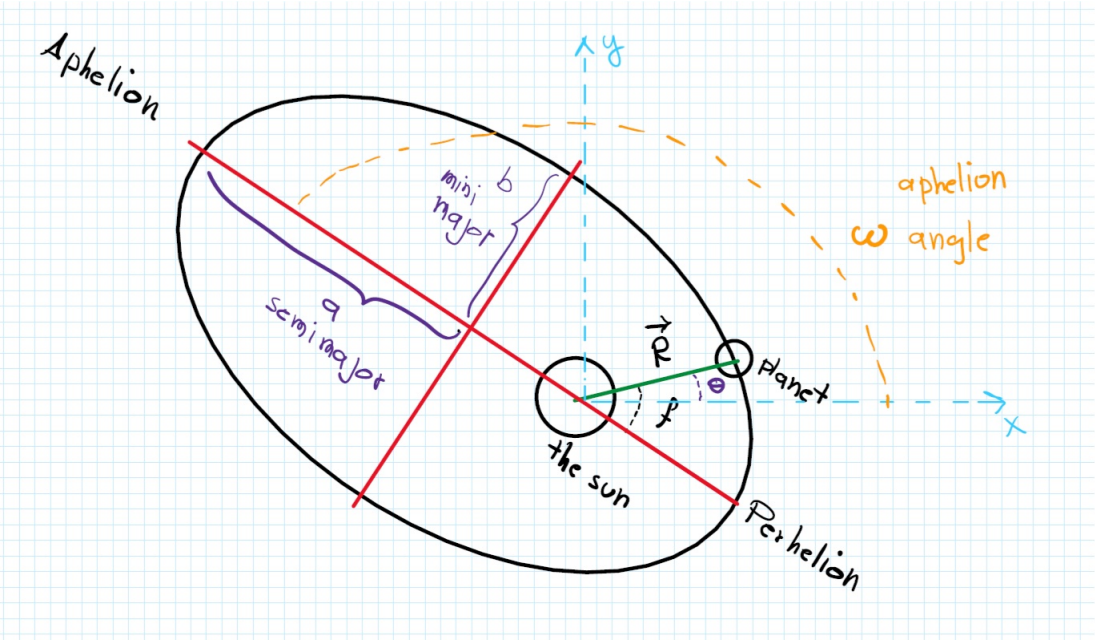


Figure 1: A sketch shows the parameters of an ellipse

Using polar coordinates , we write the components of the position vector $\vec{r}$:

- $x = r\cos\theta = r\cos(f+\omega)$
- $y = r\sin\theta = r\sin(f+\omega)$

We will define the fixed coordinate system mentioned above as the star is in its origin .which makes the star coordinates : $(0,0)$.

The numerical solution:

As we mentioned above, the numerical approach will be based on Euler-Cromer method. General form of Euler-Cromer method uses three sets of equations :

$$\begin{aligned} f''(x) &= S(x) \\ f'(x_{n+1}) &= f'(x_n) + S(x_n)\Delta x \\ f(x_{n+1}) &= f(x_n) + f'(x_{n+1})\Delta x \end{aligned} \tag{2}$$

where:

$$\begin{aligned} f''(x) &= \frac{d^2 f(x)}{dx^2}, \\ f'(x) &= \frac{df(x)}{dx}. \end{aligned} \tag{3}$$

By now it is obvious that the velocity is the first derivative of the position :

$$\vec{v}(t) = \frac{df(\vec{r})}{dt}$$

and the acceleration is the second derivative of the position vector:

$$\vec{a}(t) = \frac{df^2(\vec{r})}{dt^2}$$

where :

$$\vec{r}(t) = x(t)\hat{x} + y(t)\hat{y}$$

. In polar coordinates system will be expressed as :

$$\vec{r}(t) = r\cos\theta\hat{x} + r\sin\theta\hat{y}$$

We need to find an expression for the acceleration. We know from the law of gravity that :

$$\vec{F} = -G\frac{Mm}{r^3}\vec{r}$$

and we have from Newton second law: $\vec{F} = m\vec{a}$ By combining them together , we get:

$$m\vec{a} = -G\frac{Mm}{r^3}\vec{r} \implies \vec{a} = -G\frac{M}{r^3}\vec{r} \tag{4}$$

Inserting this expression into the the general form of Euler-Cromer:

$$\vec{r}''(t) = -G\frac{M}{r^3}\vec{r}$$
$$\vec{r}'(t_{n+1}) = \vec{r}'(t_n) + \vec{r}''(t)\Delta t$$
$$\vec{r}(t_{n+1}) = \vec{r}(t_n) + \vec{r}'(t_{n+1})\Delta t$$

Now , we have all wee need to simulate our solar system planets orbits. We choose the time to be measured in years .

3 Results

My solar system consists of 7 planets . 5 rocky planets and 2 gas revolving around a mother star with a mass $7.989710375255283 \times 10^{30}$ Kg , and a radius : 2070487.2730332096 AU. With the all other information needed are imported from"ast2000tools" package . All information are in the table 1

planet	Type	Mass[Kg]	Semi major[AU]	Eccentricity	Ω [Rad]	Period [Years]
0	Rock	1.90×10^{25}	6.6563	0.0003	-3.1416	8.5672
1	Rock	1.439×10^{25}	8.9822	0.0021	-1.5174	13.4297
2	Gas	3.35×10^{28}	55.0813	0.0557	1.7006	203.9392
3	Rock	4.08×10^{23}	4.4704	0.0519	1.0805	4.7154
4	Gas	8.58×10^{26}	26.5600	0.0314	-0.1035	68.2867
5	Rock	3.12×10^{24}	20.5532	0.0627	1.5375	46.4853
6	Rock	5.78×10^{23}	12.3234	0.0367	-1.6018	21.5820

Table 1: The information of my Solar System

By applying the equation 1 in a Python code, the plot of the analytical method was obtained ,as shown in Figure 2

Using the informations in Table 1 and inserting them into a Python code based on Euler-Cromer method with form in equation 2, we got the numerical plot shown in figure: 3

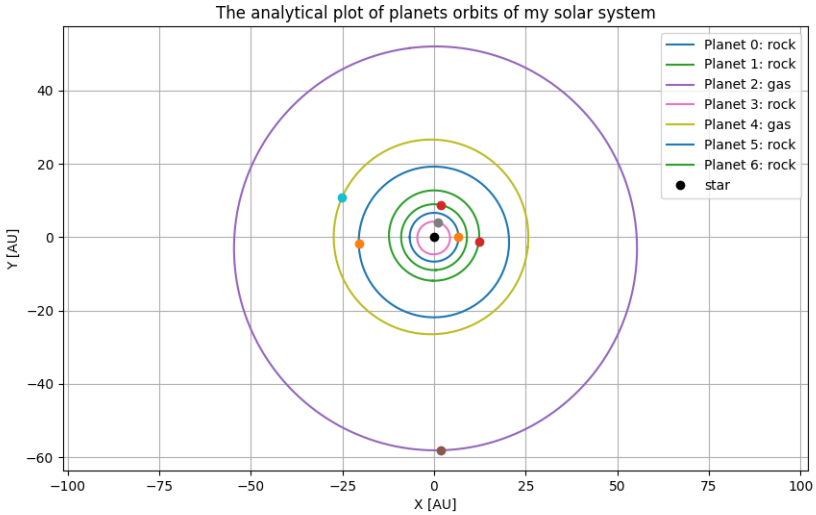


Figure 2: The Analytical Plot of 7 planets solar system with type of each planet.

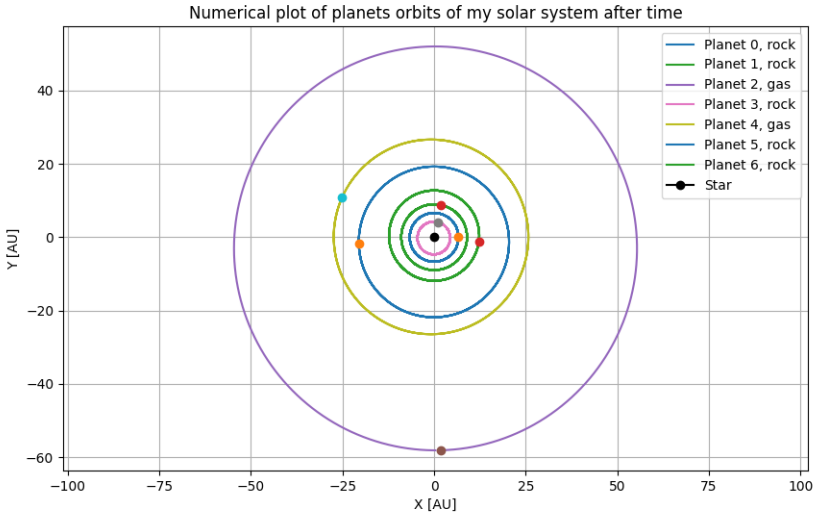


Figure 3: The Numerical Plot of 7 planets solar system with type of each planet

To check if there is any deviation between the two plots from the numerical and analytical methods, it is a good idea to combine them into one plot, as shown in the figure 4

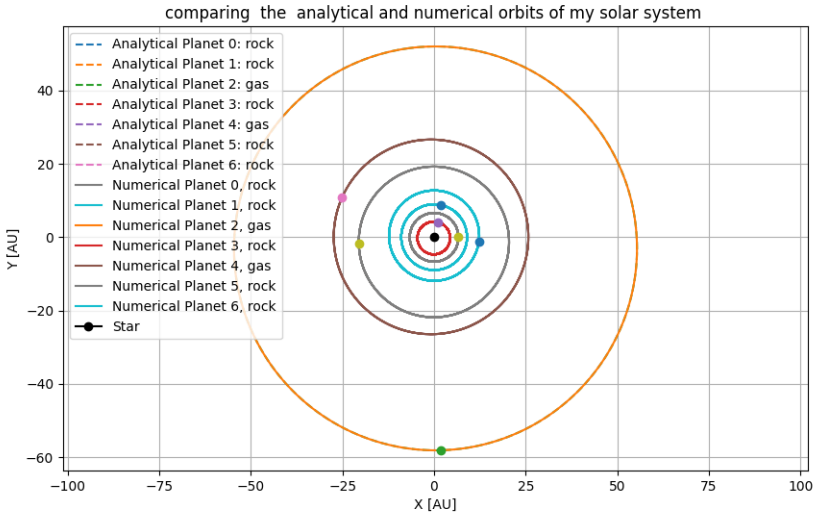


Figure 4: The combined plot for the numerical and analytical methods of 7 planets solar system with the type of each planet

4 Discussion

By comparing the plots for the analytical and the numerical plots, we can easily notice the similarity with no deviation . Our Solar System has two gas planets the third and the fifth,and the rest are rocky. Our simulation went through series of simplifications, We ignored the influence of the the planets on the star and assumed that the star is fixed in its place . In real world, The planets assert a force on star even if it is too weak, but this can give a more precision in results if we included it. Another simplification is neglecting the forces between planets which if we considered it, the model would be more close to reality.

The weakness in this model, in addition to simplifications,we used Euler-Cromer method for numerical solution.This method has some disadvantages, it does not have the accuracy when compared to other numerical methods

especially when the conservation of energy is needed. Also We did not consider the irregularities of the orbits of the first planet to the sun which has not a stable orbit as we know form Mercury in our solar system.In fact , all planets have deviation in their orbits, but it is not noticeable. In case of the inner planet, since it revolve around the sun with a high velocity in short orbit this deviation will be significant and relativistic effects can play a role in its behaviour .

5 Conclusion

Looking at our simulation, we see that it gives a good starting point.the results are logical for a simple model. When we compare the analytical and numerical plots , they look similar. We made some simple choices, like keeping the star fixed in its position and not cinsidering about how planets influence each other. We used the Euler-Cromer method, which is not be the best choice for our situation. And, we didn't consider about how some inner planet behave.

References

[1] Frode Kristian Hansen. Lecture Notes 1B *Introduction to Astrophysics,AST2000.*